

Pranay Meshram

Buffalo, NY

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Education

University at Buffalo, The State University of New York

PhD in Computer Science and Engineering

Present

University at Buffalo, The State University of New York

Master of Science: Engineering Science Robotics

Dec 2022

Technical Skills

Programming Languages: C, C++, Python, Rust

Frameworks: Numpy, OpenCV, Pandas, Sci-kit learn, Pytorch, Open3d, Gazebo, Isaac Sim, Matplotlib, Plotly, g2o, ROS

Domains: Reinforcement Learning, Computer Vision, 3D Reconstruction, Depth Estimation, Machine Learning, Geometric Optimization

Research Experience

Meta, Reality Labs

Research Scientist Intern

Redmond, WA

May 2024 - Sept 2024

Self-supervised Stereo Depth Estimation from Polarization

- Built a self-supervised stereo depth pipeline leveraging polarization physics and feature representations to enhance geometric consistency in texture-less and reflective regions.
- Designed a multi-objective learning framework jointly predicting depth and surface normals, achieving a 20% accuracy improvement over state-of-the-art stereo models.

University at Buffalo, The State University of New York

Research Assistant

Present

HOPHY: Hierarchical Off-Road Planning using Hypergraphs

- Modeled a learning-based cost function for traversal risk and time prediction, enabling few-shot tuning that reduced training time by 80% (using only 20% training data) when adapting a PPO-based Deep Reinforcement Learning locomotion policy for a Unitree Go1 in NVIDIA Isaac Sim, bridging the sim-to-real gap for robust terrain traversal.
- Developed HOPHY, compressing terrain into three semantic levels (fine, geometric-semantic, coarse) to reduce millions of pixels to hundreds of planning units.
- Modeled terrain constraints as hyperedges enabling $O(|E_{hyper}|)$ cost updates under dynamic environments (rain, snow) without recomputing the spatial graph.
- Integrated with Genetic Algorithm task allocation to enable real-time replanning, achieving $2836\times$ speedup over A* and $22\times$ over flat semantics on kilometer-scale maps.

CLEAR: Semantic-Geometric Terrain Abstraction

- Developed CLEAR (Connected Landcover Elevation Abstract Representation), fusing dense elevation and semantic land-cover data to accurately estimate physics-based traversability, enabling large-scale autonomous navigation.
- Demonstrated $10\times$ faster planning with only 6.7% cost overhead, yielding 6–9% shorter and significantly more reliable paths compared to existing grid and mesh-based baseline abstractions.

Quality-Aware 3D Reconstruction

- Designed Quality-Aware Loss (QAL) decoupling precision and recall, improving coverage by +4.3 points and recovering fine structures.
- Created a 4D quality metric for point-clouds comprehensively measuring 3D reconstruction coverage, precision, and consistency.

Efficient Deep Learning for Edge Perception

- Designed lightweight CNN architectures optimized for FPGA and embedded GPUs, enabling on-device vision inference.
- Reduced latency and memory footprint through quantization-aware training and custom kernel fusion.

Publications

- QAL: A Loss for Recall-Precision Balance in 3D Reconstruction. Accepted in WACV 2026.
- HOPHY: Hierarchical Off-Road Planning using Hypergraphs. Under submission RSS 2026.
- CLEAR: A Semantic-Geometric Terrain Abstraction for Large-Scale Unstructured Environments. Under review RA-L.
- Empir3D: Point-Cloud Quality Assessment Framework. Under Revision.

Professional Experience

NVIDIA Graphics Pvt Ltd

System Software Engineer

Pune, India

Dec 2018 – May 2021

- Led development of CUDA language program generating Fuzzer using Deep Learning and implemented solutions reducing engineering effort by 30%.
- Managed the design and verification of the CUDA Compiler for Automotive Safety Project to satisfy ISO 26262.

Renishaw Metrology Systems Ltd

Software Engineer

Pune, India

Jul 2017 – Dec 2018

- Implemented voxel-based motion path planning and probe compensation algorithms for precision metrology.
- Applied Particle Swarm Optimization to calibration routines → 25% reduction in recurring bug count.

Accomplishments and Achievements

2022: DAC System Design Contest 2022 – 4th place for object detection model on an FPGA.

2022: ACM/IEEE TinyML Design Contest at ICCAD – 1st place for binary classification model on an ultra-low power device.